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Department of Computer Sciences

End of Studies Project

In view of obtaining the
Applied Bachelor's Degree in Computer Science
Specialty : Information Systems Development

Study and Development of a Smart Training Platform "ADVANCIA"

A multi-actor web platform for online training management

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Internship Company: ADVANCIA TRAINING CENTER

— Academic Year 2025 / 2026 —

Dedication



*To my beloved parents,
the unwavering pillars of my life, whose endless love,
patience and sacrifices have shaped the man I am today.
No words can ever capture the depth of my gratitude.*

*To my brothers and sisters,
my first companions, my safe harbor in every storm; thank
you for the laughter, the encouragement and the silent
strength you have given me throughout this journey.*

*To my dear friends,
those who walked beside me during the long nights of work
and the brightest moments of joy; your loyalty has been a
true blessing.*

*To my teachers and supervisors,
who lit my path with knowledge and guided my steps with
rigor and kindness; this work is the fruit of your trust.*

*To everyone
who believed in me, even briefly, I dedicate this modest
achievement.*

Sayhi MAHDI

Dedication



*To my dear mother,
whose tenderness and prayers have always been the light-
house of my life. Every line of this work carries a fragment
of your love.*

*To my beloved father,
the role model of patience, dignity and hard work. Your
trust gave me the courage to dream, and your wisdom
taught me how to build.*

*To my brothers and sisters,
my forever support and my dearest companions; thank you
for cheering on every small victory and for standing close
in every difficult moment.*

*To my closest friends,
for the laughter shared, for the late-night encouragements,
and for being a second family along this academic journey.*

*To my professors at ISET Siliana,
who shaped not only my technical skills but also my way of
thinking; your dedication is engraved in this work.*

*To every soul
who, by a kind word or a sincere wish, helped me reach this*

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The Authors

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List of Acronyms

Acronym	Signification
API	Application Programming Interface
BDD	Base De Données
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DGET	Direction Générale des Études Technologiques
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
ISSET	Institut Supérieur des Études Technologiques
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model – View – Controller
ORM	Object Relational Mapping
PDF	Portable Document Format
PFE	Projet de Fin d'Études
REST	REpresentational State Transfer
SCRUM	Iterative agile development framework
SGBD	Système de Gestion de Base de Données
SPA	Single Page Application
SQL	Structured Query Language
SSL	Secure Sockets Layer
UI / UX	User Interface / User Experience
UML	Unified Modeling Language
URL	Uniform Resource Locator

General Introduction

The digital revolution has redrawn the boundaries of how knowledge is produced, accessed and exchanged. In a few short years, the act of *learning* has shifted from an exclusively physical experience – bound by the walls of a classroom – towards a flexible, connected and increasingly personalized journey, available anywhere and at any time. This silent transformation has placed online training at the very heart of professional growth, lifelong learning and competitiveness in the labor market.

In Tunisia, training centers and technological institutes are at a turning point. The expectations of learners have evolved : they no longer seek a static catalog of courses, but a smart, interactive and personalized environment, capable of guiding them, tracking their progress and helping them make the right choices. At the same time, training organizations need powerful management tools to handle their users, their offerings, their reservations and their administrative workflows in a fluid and reliable way.

It is in this context that our end-of-studies project takes place, carried out within **ADVANCIA Training Center** as part of our Applied Bachelor's Degree in Computer Science at the **Institut Supérieur des Études Technologiques de Siliana**. The mission entrusted to us was clear in its ambition : to design and develop a **smart web platform for training management**, named **ADVANCIA**, capable of bringing together *users, trainers, administrators* and *super-administrators* within a single, secure and modern ecosystem.

Our solution is built around four well-defined actors. The **User** can create an account, browse the available trainings, reserve sessions, manage his profile and follow his personal dashboard. The **Trainer** animates the courses he is responsible for and follows his learners. The **Administrator** pilots the platform on a daily basis : managing users, validating reservations, importing and exporting data in Excel or PDF, and exploiting a smart dashboard supported by an integrated chatbot. Finally, the **Super-Administrator** holds the highest level of authority, supervising every actor and every operation of the system.

To structure this work in a coherent and progressive manner, the present dissertation is organized into six chapters that reflect the methodological journey we followed.

- **Chapter 1 – Project Framework Presentation :** introduces the host company ADVANCIA TRAINING CENTER, situates the project in its context, identifies the problem and presents the proposed solution along with the chosen methodology.
- **Chapter 2 – Analysis and Project Planning :** captures the functional and non-functional requirements, identifies the actors, formalizes the global use case diagram, and structures the work into releases and sprints supported by a Gantt schedule.
- **Chapter 3 – Release 1 :** addresses the authentication and user-management module, the foundation of the security and identity layer of the platform.
- **Chapter 4 – Release 2 :** details the training catalog and the course-management module, which form the functional heart of the platform.
- **Chapter 5 – Release 3 :** covers the reservation system, the notifications, the smart dashboard and the integrated chatbot.
- **Chapter 6 – Realization :** presents the technical environment, the architecture, the development tools and the most representative interfaces of the final product.

A general conclusion closes this dissertation, summarizing the contributions of the work, evaluating its results against the initial objectives and tracing concrete perspectives of evolution for the platform.

Chapter 1 Project Framework Presentation

1. Introduction
2. Host Company Presentation
3. Project Context and Problem Statement
4. Study of the Existing Solutions
5. Proposed Solution
6. Adopted Methodology : Scrum
7. Conclusion

1.1 Introduction

This first chapter is dedicated to setting the general framework of our end-of-studies project. We begin by presenting the host company ADVANCIA TRAINING CENTER, then we outline the context and the problem the project is meant to solve. After a critical study of the existing solutions, we describe our proposed platform and justify the choice of the agile *Scrum* methodology that has guided every stage of our work.

1.2 Host Company Presentation

1.2.1 Identity

ADVANCIA TRAINING CENTER is a Tunisian training organization specialized in continuous education and professional certification in the fields of computer science, networks, management and personal development. The center accompanies students, professionals and companies in their up-skilling journey through tailored programs delivered both *on-site* and *online*.

1.2.2 Information sheet

Field	Description
Name	ADVANCIA Training Center
Sector of activity	Professional training and certifications
Domains	IT, Networks, Management, Soft Skills
Target audience	Students, professionals, companies
Training modes	In-person, Online, Hybrid
Address	Tunis, Tunisia
Website	https://www.advancia-tc.com

Table 1.1: Identity sheet of ADVANCIA Training Center

1.2.3 Organizational chart

The figure below summarizes the simplified organizational structure of the host company.

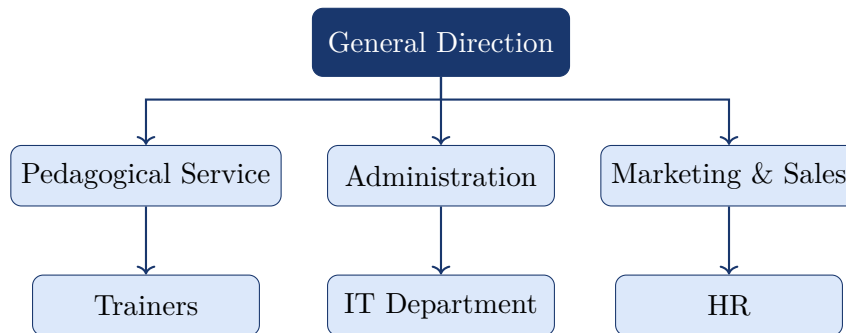


Figure 1.1: Organizational chart of ADVANCIA Training Center

1.3 Project Context and Problem Statement

1.3.1 Context

This project takes place within the framework of an end-of-studies internship in computer science, in collaboration between **ISSET Siliana** and **ADVANCIA Training Center**. It aims at modernizing the way the company handles its trainings and interacts with its users, by replacing manual or scattered processes with a centralized and intelligent web platform.

1.3.2 Problem statement

After studying the company's daily workflow, we identified several recurring difficulties that hinder both the user experience and the administrative efficiency :

- **Lack of a unified digital channel** : reservations are taken by phone, e-mail or in person, generating ambiguity and loss of information.
- **Manual user management** : user records and registrations are handled through spreadsheets, which is error-prone and not scalable.
- **No real-time visibility** : the management has no smart dashboard to track users, reservations or revenue.

- **Limited communication** : no automatic notifications when a reservation is accepted, refused or modified.
- **No integrated assistance** : users have no immediate way to ask for help, beyond classical e-mail support.

1.3.3 Synthetic statistics on the existing situation

To quantify the problem, we conducted a small internal survey within the host company. The chart below summarizes the main pain points reported by the staff.

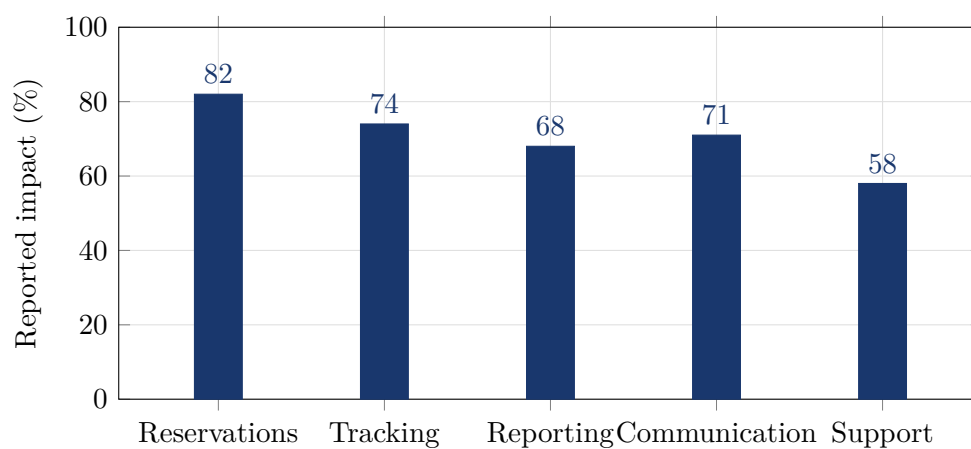


Figure 1.2: Pain points reported by the staff (internal survey, $n = 12$)

1.4 Study of the Existing Solutions

Several training-management platforms exist on the market. We compared three of the most representative ones to identify what is missing and what we can do better.

Criteria	Moodle	Udemy Business	Coursera
Multi-actor management	Partial	Limited	Limited
Smart dashboard / KPIs	No	Partial	Partial
Integrated chatbot	No	No	No
Reservation workflow	No	No	No
Excel / PDF export	Partial	No	No
Local deployment cost	Free / Self-hosted	Paid	Paid

Table 1.2: Comparative study of existing solutions

1.4.1 Critique

The studied platforms are powerful for content delivery, but they remain generic. None of them provides a complete *four-actor reservation workflow*, a *smart dashboard with chatbot* and a *multi-format export* (Excel/PDF) tailored for a Tunisian training center.

1.5 Proposed Solution

We propose **ADVANCIA**, a smart, secure and responsive web platform that brings together every actor of the training process inside one ecosystem. The figure below summarizes the high-level functional architecture of the proposed solution.

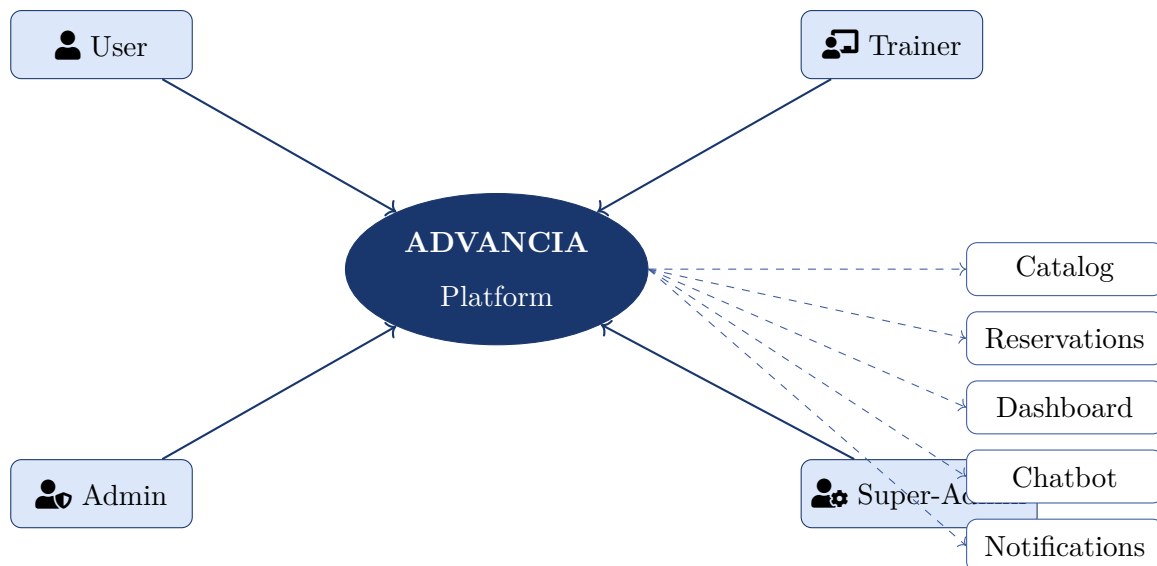


Figure 1.3: High-level overview of the ADVANCIA platform

1.6 Adopted Methodology : Scrum

1.6.1 Why agile ?

The functional perimeter of *ADVANCIA* is broad and the requirements are likely to evolve. A **rigid waterfall** approach would have led to a tunnel effect : a long phase of specification, then a long development, then a single deliverable – with all the risks that this implies. We therefore chose the **Scrum** agile framework, which is iterative, incremental and oriented around tangible value at the end of each cycle.

1.6.2 Scrum overview

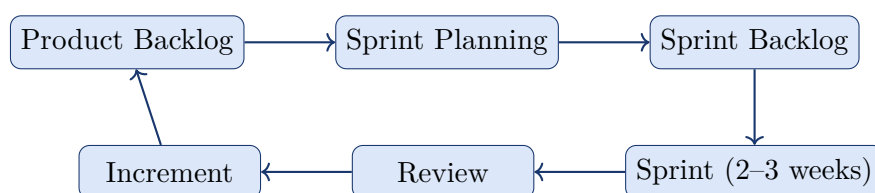


Figure 1.4: The Scrum cycle adopted in this project

1.6.3 Scrum roles in our project

Role	Holder & mission
Product Owner	Professional supervisor at ADVANCIA – defines the vision and prioritizes the backlog.
Scrum Master	Academic supervisor at ISET Siliana – enforces the process and removes blockers.
Development Team	Istabrak BEN KHMAYS & Sayhi MAHDI – design, develop, test and deliver the increments.

Table 1.3: Scrum roles in the ADVANCIA project

1.7 Conclusion

This first chapter has settled the foundations of our work : the host company, the context, the limits of existing solutions, the proposed platform and the agile process that will drive the implementation. The next chapter goes deeper, by capturing the full set of requirements and by translating them into a structured plan of releases and sprints.

Chapter 2 Analysis and Project Planning

1. Introduction
2. Identification of the Actors
3. Functional Requirements
4. Non-Functional Requirements
5. Global Use Case Diagram
6. Product Backlog
7. Releases and Sprints
8. Provisional Schedule (Gantt)
9. Conclusion

2.1 Introduction

After framing the project, this chapter focuses on understanding *what* the ADVANCIA platform must do and *how* the work is organized. We start by identifying every actor of the system, we then formalize the functional and non-functional requirements, we draw the global use case diagram and finally we transform the whole into a structured plan of releases, sprints and a Gantt schedule.

2.2 Identification of the Actors

The ADVANCIA platform is a multi-actor system. Four main actors interact with the system, each one with a clearly defined scope of responsibilities.

Actor	Role and responsibilities
User	Creates and manages his account, browses and reserves trainings, manages his profile (photo, avatar, password) and consults his personal dashboard.
Trainer	Handles the trainings he is responsible for, follows enrolled users, animates sessions and updates pedagogical content.
Administrator	Manages users, validates or refuses reservations, imports and exports data (Excel, PDF), exploits a smart dashboard and an integrated chatbot.
Super-Administrator	Has the highest level of authority : manages users, admins, trainers, trainings and courses, sends notifications and supervises the whole platform.

Table 2.1: Actors of the ADVANCIA platform

2.3 Functional Requirements

The functional requirements describe *what* the system must offer. They are grouped by actor.

2.3.1 User

- Create an account and authenticate.
- Manage the personal profile : photo, avatar, password, contact info.
- Browse and search the catalog of trainings.
- Reserve a training session.
- Consult a simple personal dashboard with the essential indicators.
- Receive notifications about reservations and account events.

2.3.2 Trainer

- Authenticate and manage profile.
- Consult the list of trainings and sessions assigned to him.
- Track enrolled users.
- Update pedagogical resources of his courses.

2.3.3 Administrator

- Manage users (create, update, suspend, delete).
- Import data (e.g. CSV/Excel of users) and export data in Excel or PDF.
- Use the smart dashboard with KPIs and charts.
- Use the integrated chatbot for assistance and quick queries.
- Manage personal profile.

2.3.4 Super-Administrator

- All capabilities of the Administrator.
- Manage admins and trainers.
- Manage trainings and courses (CRUD).
- Accept or refuse reservations.

- Send notifications to users or administrators.
- Change password and supervise security policies.

2.4 Non-Functional Requirements

Requirement	Description
Security	Authentication via JWT, password hashing, role-based access control, HTTPS.
Performance	Response time below 2 seconds for the main pages, smooth navigation.
Usability	Modern, clear and responsive UI accessible from desktop, tablet and mobile.
Reliability	Daily database backup and graceful error handling.
Scalability	Modular architecture allowing the addition of new modules without disruption.
Maintainability	Clean code, documented modules, version-controlled with Git.
Portability	Cross-browser compatibility (Chrome, Firefox, Edge).

Table 2.2: Non-functional requirements

2.5 Global Use Case Diagram

The following diagram synthesizes the main use cases of the ADVANCIA platform and the relationships between actors and use cases.

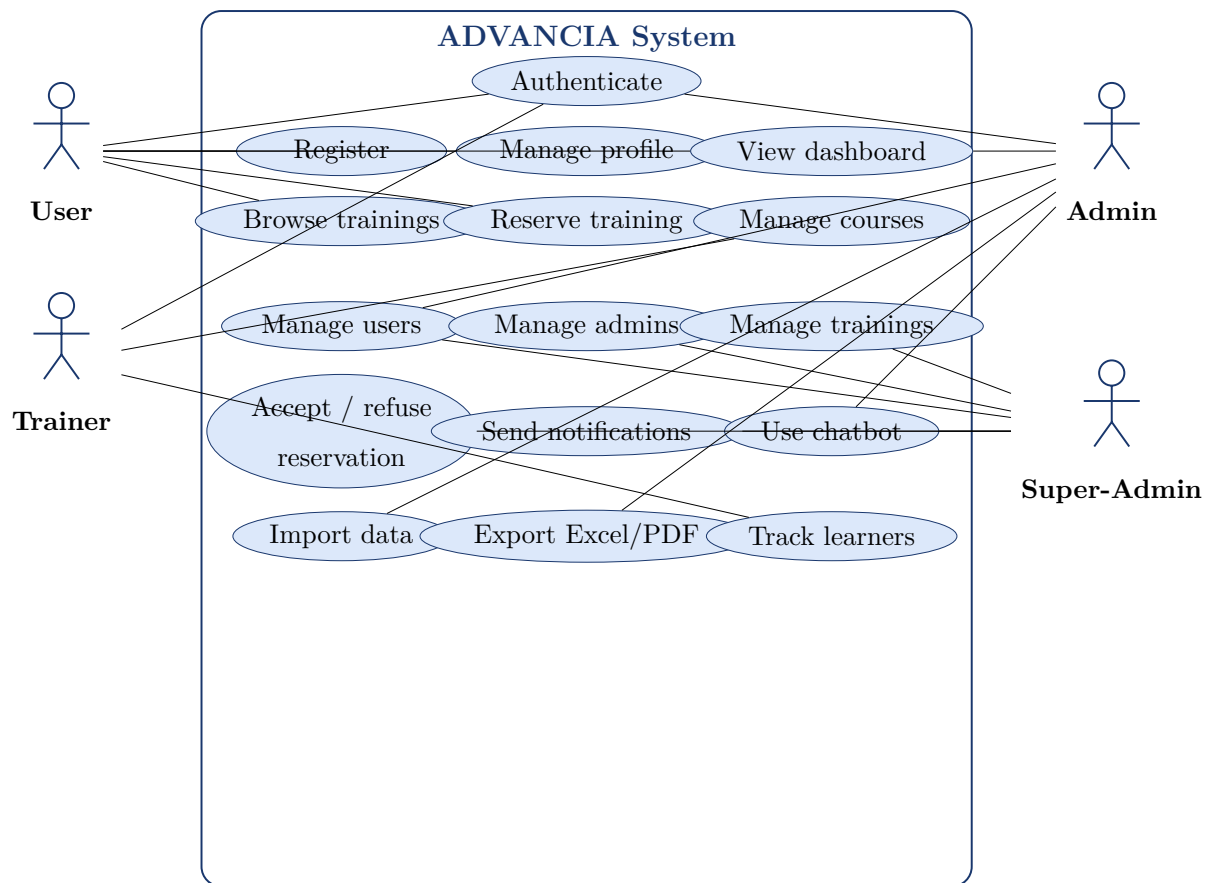


Figure 2.1: Global use case diagram of the ADVANCIA platform

2.6 Product Backlog

The product backlog gathers, in priority order, every user story to be developed. The story points (SP) follow a Fibonacci scale.

ID	User Story	Actor	SP
US01	As a user, I want to create an account so I can access the platform.	User	3
US02	As a user, I want to authenticate securely.	User	3
US03	As a user, I want to manage my profile (photo, avatar, password).	User	5
US04	As a user, I want a personal dashboard.	User	5
US05	As a user, I want to browse and search trainings.	User	5
US06	As a user, I want to reserve a training session.	User	5

ID	User Story	Actor	SP
US07	As a trainer, I want to consult my assigned trainings.	Trainer	3
US08	As a trainer, I want to track my enrolled learners.	Trainer	5
US09	As an admin, I want to manage users (CRUD).	Admin	8
US10	As an admin, I want to import data from Excel.	Admin	5
US11	As an admin, I want to export data to Excel and PDF.	Admin	5
US12	As an admin, I want a smart dashboard.	Admin	8
US13	As an admin, I want an integrated chatbot.	Admin	8
US14	As a super-admin, I want to manage admins and trainers.	Super-Admin	8
US15	As a super-admin, I want to manage trainings and courses.	Super-Admin	8
US16	As a super-admin, I want to accept or refuse reservations.	Super-Admin	5
US17	As a super-admin, I want to send notifications.	Super-Admin	5
US18	As any actor, I want a responsive and modern UI.	All	5

Table 2.3: Product backlog of the ADVANCIA platform

2.7 Releases and Sprints

The product backlog has been broken down into **three releases**, each one delivering a tangible set of value at the end of its lifecycle.

Release	Scope	Sprints
Release 1	Authentication & User management (US01–US04, US18)	S1, S2
Release 2	Training catalog & Course management (US05, US07, US15)	S3, S4
Release 3	Reservation, Notifications, Dashboard & Chatbot (US06, US08–US14, US16, US17)	S5, S6

Table 2.4: Decomposition into releases and sprints

2.7.1 Distribution of effort

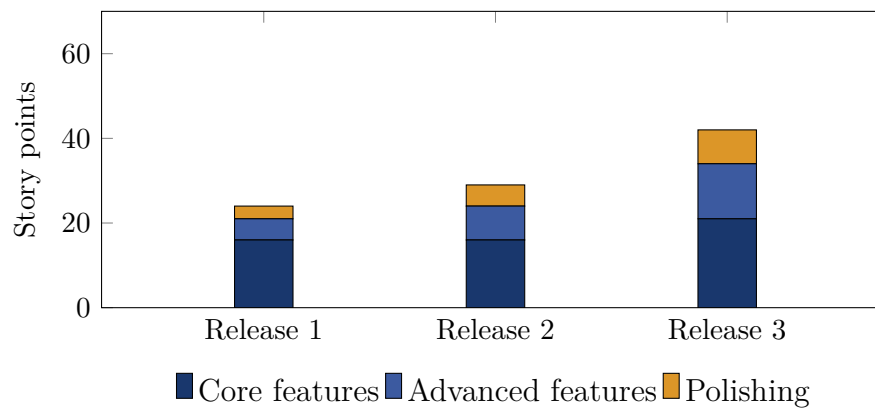


Figure 2.2: Distribution of story points across releases

2.8 Provisional Schedule (Gantt Chart)

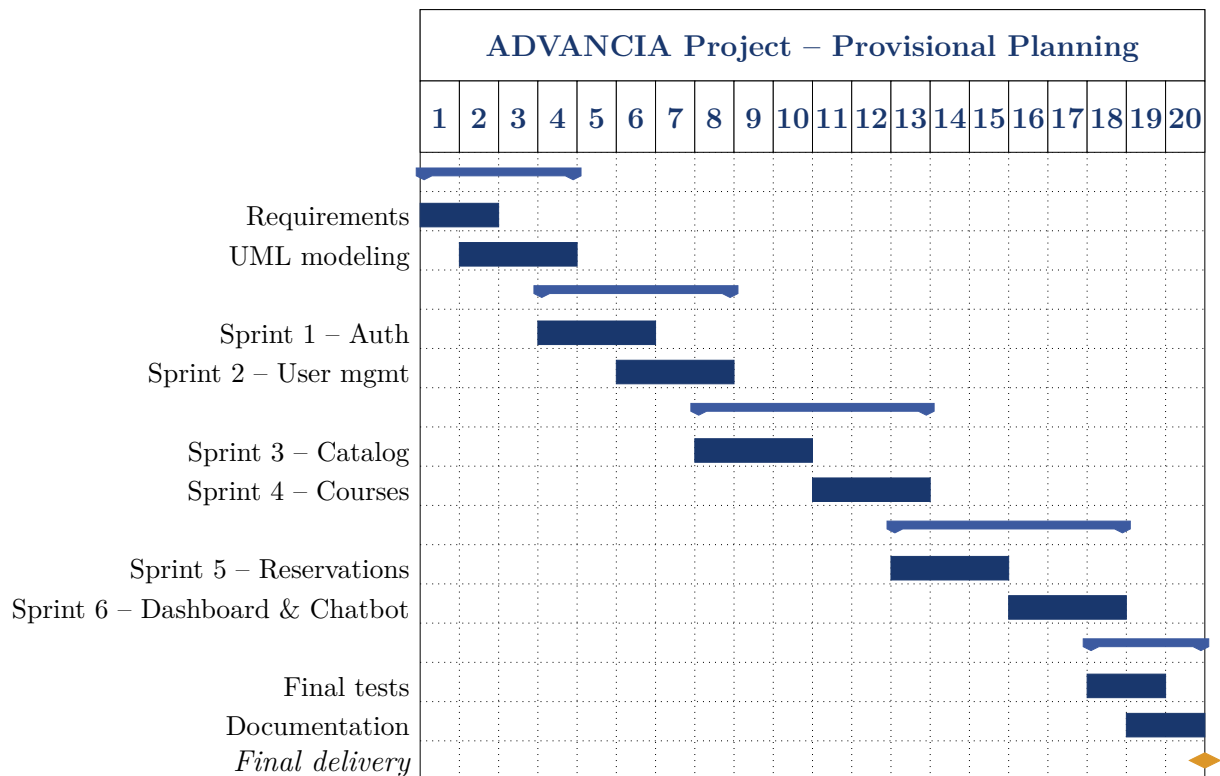


Figure 2.3: Gantt chart of the ADVANCIA project

2.9 Conclusion

This second chapter has clarified the perimeter of the platform : its four actors, its functional and non-functional requirements, the global use cases, and the structuring of the work into three releases driven by a Gantt schedule. The next chapters will detail each release, sprint by sprint, until the final integrated solution.

Chapter 3 Release 1 : Authentication & User Management

1. Introduction
2. Sprint Backlog
3. Refined Use Cases
4. Sequence Diagrams
5. Class Diagram of the Module
6. Achieved Increment
7. Conclusion

3.1 Introduction

The first release lays the security foundations of the platform. It delivers everything that revolves around *identity* : registration, secure authentication, profile management and the personal user dashboard. Without this layer, no other module can be safely built on top.

3.2 Sprint Backlog

ID	Task	Sprint	Status
T1	Database design (users, roles, sessions)	S1	✓
T2	Sign-up form with field validation	S1	✓
T3	Login with JWT & password hashing	S1	✓
T4	Forgot / reset password	S1	✓
T5	User profile : photo & avatar upload	S2	✓
T6	Change password from profile	S2	✓
T7	Personal dashboard widgets	S2	✓
T8	Role-based access middleware	S2	✓

Table 3.1: Sprint backlog – Release 1

3.3 Refined Use Cases

3.3.1 Detailed use case : *Sign-up*

Title	Sign-up
Actor	User
Goal	Create a personal account on the platform.
Pre-condition	The user is not yet registered.
Main scenario	1. The user opens the sign-up page. 2. The system displays the form. 3. The user fills in the required fields. 4. The system validates the data. 5. The system creates the account and sends a confirmation.
Alternative	4.a If the e-mail already exists, the system displays an error.
Post-condition	The account is created and ready to be activated.

Table 3.2: Detailed use case : Sign-up

3.3.2 Use case diagram of Release 1

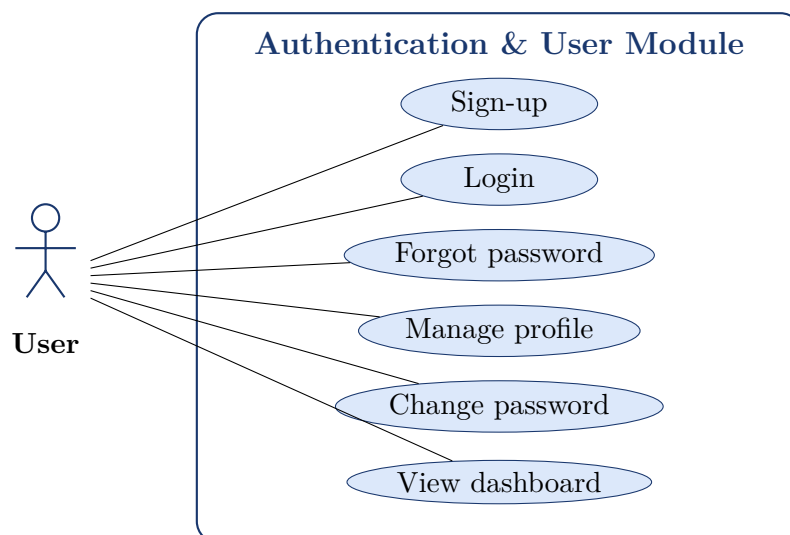


Figure 3.1: Use case diagram – Release 1

3.4 Sequence Diagrams

3.4.1 Sequence : Login

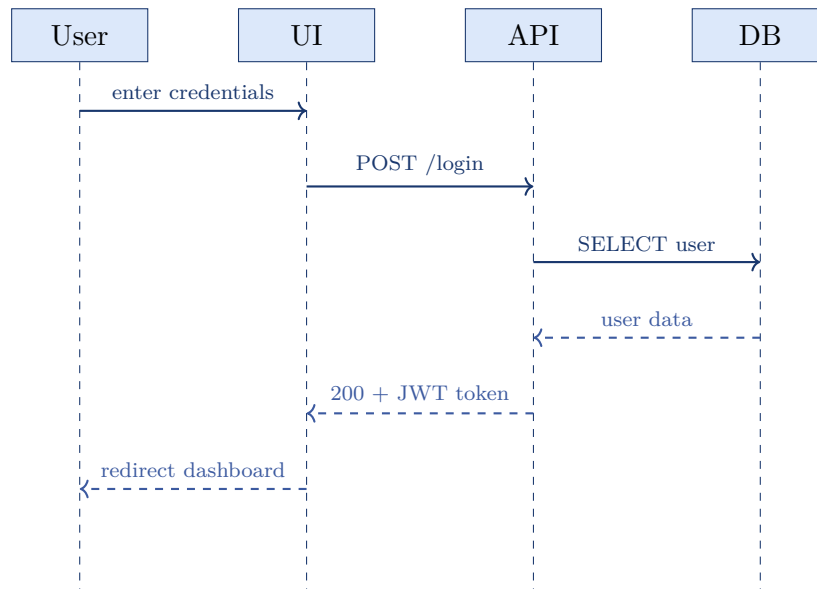


Figure 3.2: Sequence diagram of the login process

3.5 Class Diagram of the Module

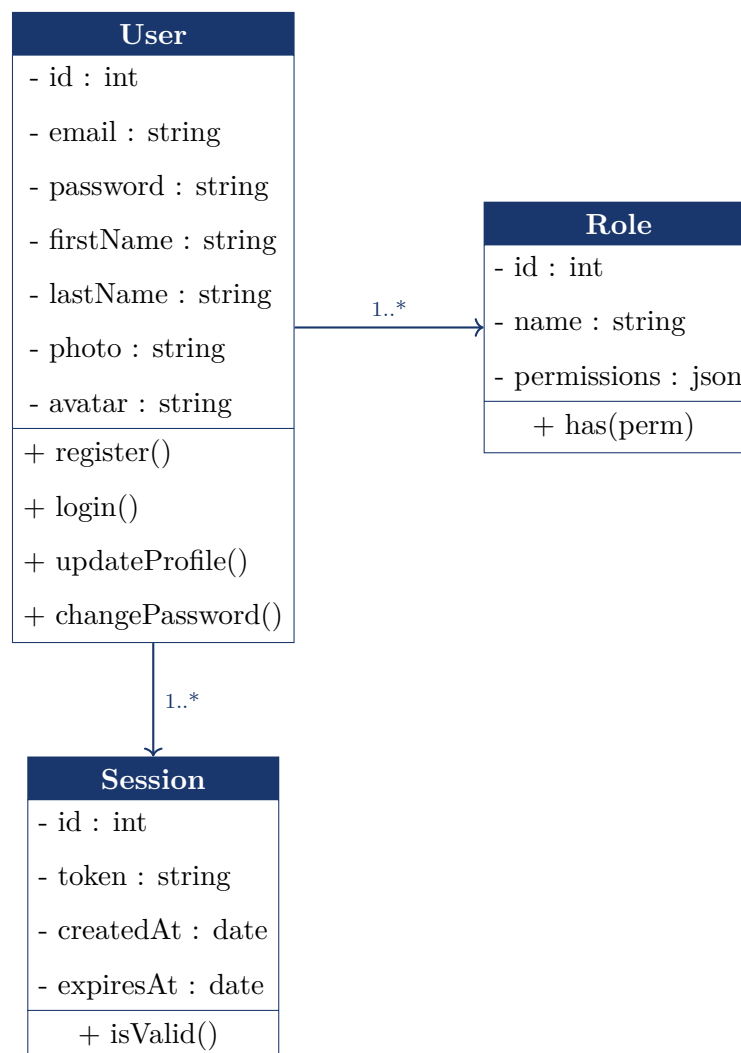


Figure 3.3: Class diagram – Authentication module

3.6 Achieved Increment

At the end of Release 1, the platform delivers :

- A complete sign-up / login system based on JWT.
- Profile management with photo and avatar upload.
- Password reset and change.
- A first version of the personal dashboard for the user.
- A role-based middleware ready to be reused by the next releases.

Indicative interfaces. The figure below positions, schematically, the screens delivered at the end of this release.

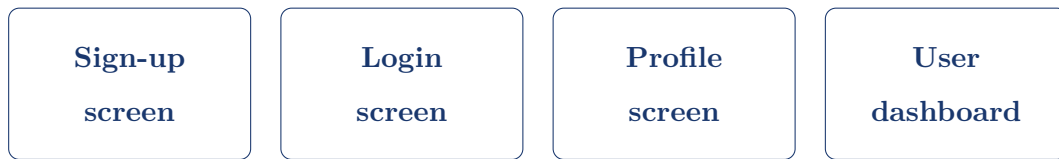


Figure 3.4: Screens delivered at the end of Release 1

3.7 Conclusion

This first release has stabilized a robust security and identity layer. The next chapter builds on top of it to offer the functional heart of the platform : the training catalog and the course-management module.

Chapter 4 Release 2 : Training Catalog & Course Management

1. Introduction
2. Sprint Backlog
3. Refined Use Cases
4. Sequence Diagrams
5. Class Diagram of the Module
6. Achieved Increment
7. Conclusion

4.1 Introduction

The second release shifts the focus from *identity* to *content*. It builds the training catalog browsed by users and the course-management workspace used by trainers and the super-administrator. This is the functional heart of the platform : without trainings, there is nothing to reserve.

4.2 Sprint Backlog

ID	Task	Sprint	Status
T9	DB design (training, course, category, trainer)	S3	✓
T10	Public catalog : list, filter, search	S3	✓
T11	Training detail page	S3	✓
T12	CRUD trainings (super-admin)	S4	✓
T13	CRUD courses inside a training (super-admin / trainer)	S4	✓
T14	Trainer assignment to a training	S4	✓
T15	Image & document upload for a course	S4	✓

Table 4.1: Sprint backlog – Release 2

4.3 Refined Use Cases

4.3.1 Detailed use case : *Browse trainings*

Title	Browse trainings
Actor	User
Goal	Discover and filter the available trainings.
Pre-condition	The user has access to the catalog page.
Main scenario	1. The user opens the catalog. 2. The system displays the list of trainings. 3. The user filters by category, level or trainer. 4. The system updates the result list dynamically. 5. The user opens a training to see its detailed page.
Alternative	2.a If the catalog is empty, the system displays a friendly empty state.
Post-condition	The user has all the information needed to decide whether to reserve.

Table 4.2: Detailed use case : Browse trainings

4.3.2 Use case diagram of Release 2

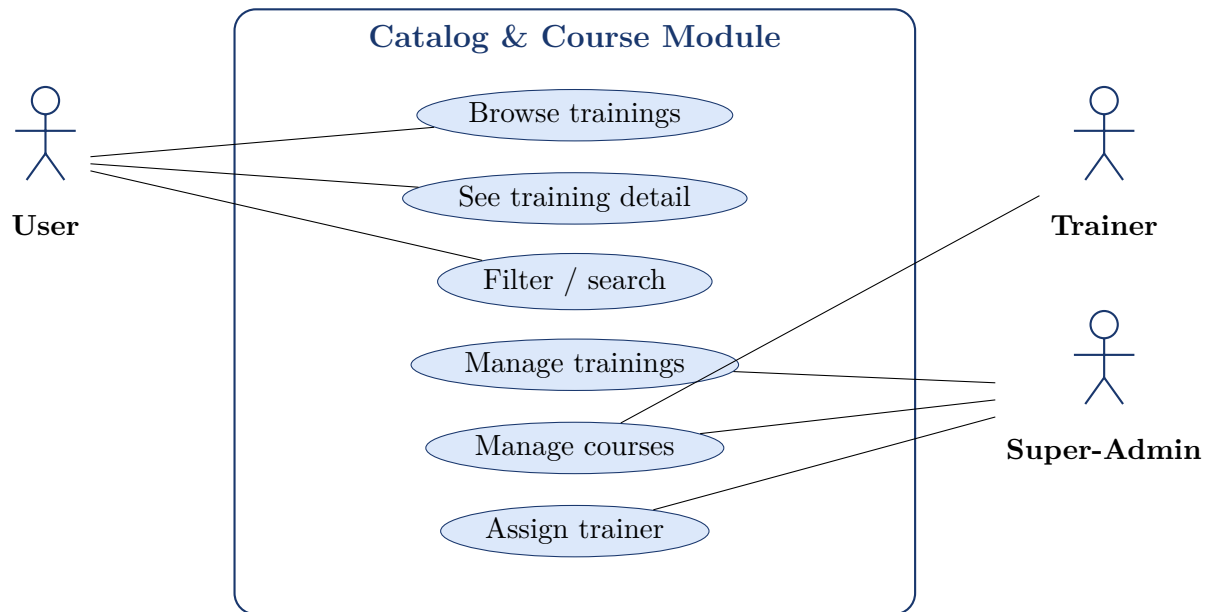


Figure 4.1: Use case diagram – Release 2

4.4 Sequence Diagrams

4.4.1 Sequence : Add a training (super-admin)

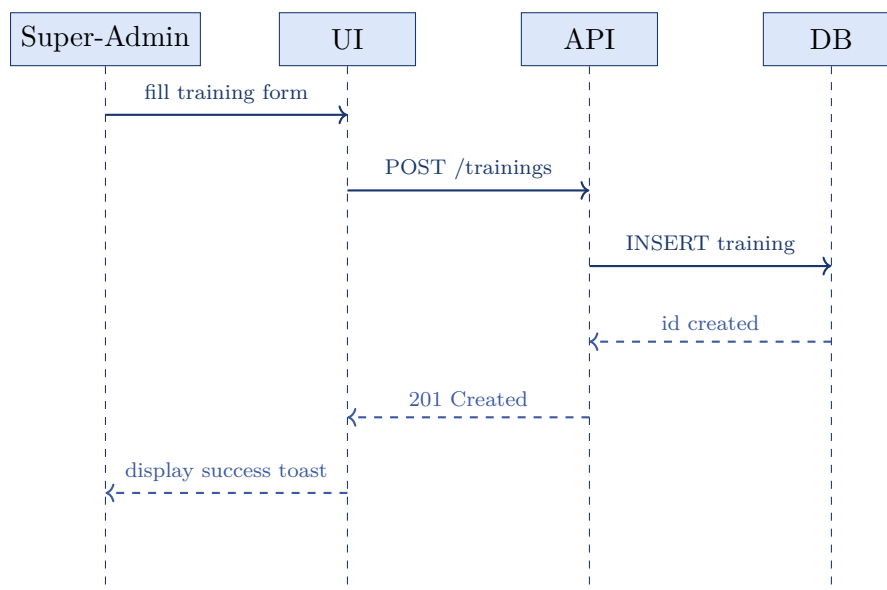


Figure 4.2: Sequence diagram : Add a training

4.5 Class Diagram of the Module

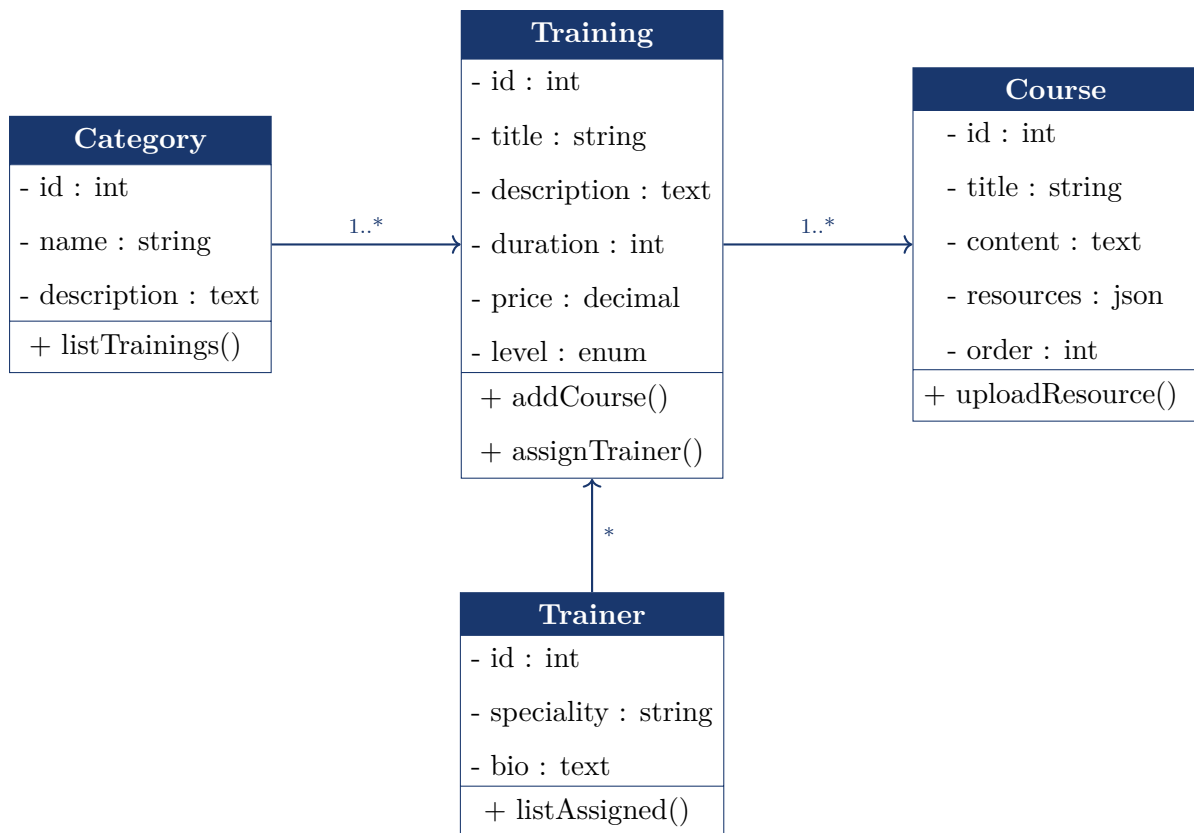


Figure 4.3: Class diagram – Catalog & Course module

4.6 Achieved Increment

- A public catalog with filter and search.
- A detailed page for each training, with the list of its courses.
- A super-admin workspace for the full CRUD on trainings, categories and courses.
- Trainer assignment, with a dedicated personal area for trainers.

4.7 Conclusion

The platform now exposes a real, browsable training catalog backed by a complete administration workspace. The third release will activate the most dynamic part of the system : reservations, notifications, the smart dashboard and the chatbot.

Chapter 5 Release 3 : Reservation, Notifications, Dashboard & Chatbot

1. Introduction
2. Sprint Backlog
3. Refined Use Cases
4. Sequence Diagrams
5. Class Diagram of the Module
6. Smart Dashboard & KPIs
7. Achieved Increment
8. Conclusion

5.1 Introduction

The third release brings the platform to life. It activates the *reservation* workflow, the *notifications* channel, the *smart dashboard* for the administrators and the *chatbot* that turns the dashboard into a conversational tool. It also closes the import / export feature for Excel and PDF.

5.2 Sprint Backlog

ID	Task	Sprint	Status
T16	DB design : reservation, notification	S5	✓
T17	Reserve a training (user)	S5	✓
T18	Accept / refuse reservation (super-admin)	S5	✓
T19	E-mail & in-app notifications	S5	✓
T20	Smart dashboard : KPIs & charts	S6	✓
T21	Chatbot integration	S6	✓
T22	Import users from Excel	S6	✓
T23	Export data to Excel & PDF	S6	✓

Table 5.1: Sprint backlog – Release 3

5.3 Refined Use Cases

5.3.1 Detailed use case : *Reserve a training*

Title	Reserve a training
Actor	User
Goal	Book a seat in an upcoming training session.
Pre-condition	The user is authenticated and the training is open for reservations.
Main scenario	1. The user opens the training detail page. 2. The user clicks <i>Reserve</i>. 3. The system creates a reservation with status <i>pending</i>. 4. The super-admin receives a notification. 5. The super-admin accepts or refuses the reservation. 6. The user receives a notification of the decision.
Alternative	5.a If refused, the reservation is closed and the seat is freed.
Post-condition	The user has a confirmed (or refused) reservation traced in his dashboard.

Table 5.2: Detailed use case : Reserve a training

5.3.2 Activity diagram : Reservation workflow

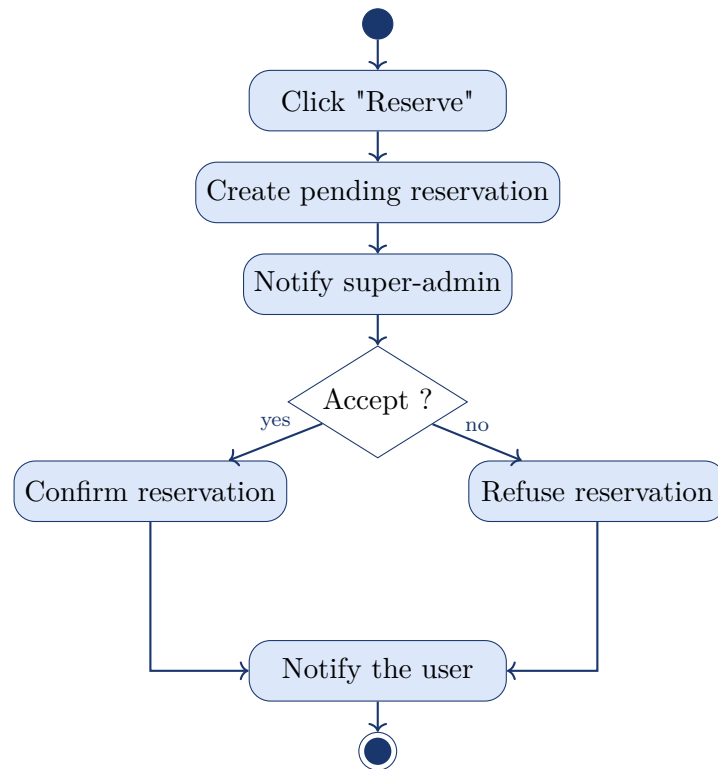


Figure 5.1: Activity diagram of the reservation workflow

5.4 Sequence Diagrams

5.4.1 Sequence : Reserve and notify

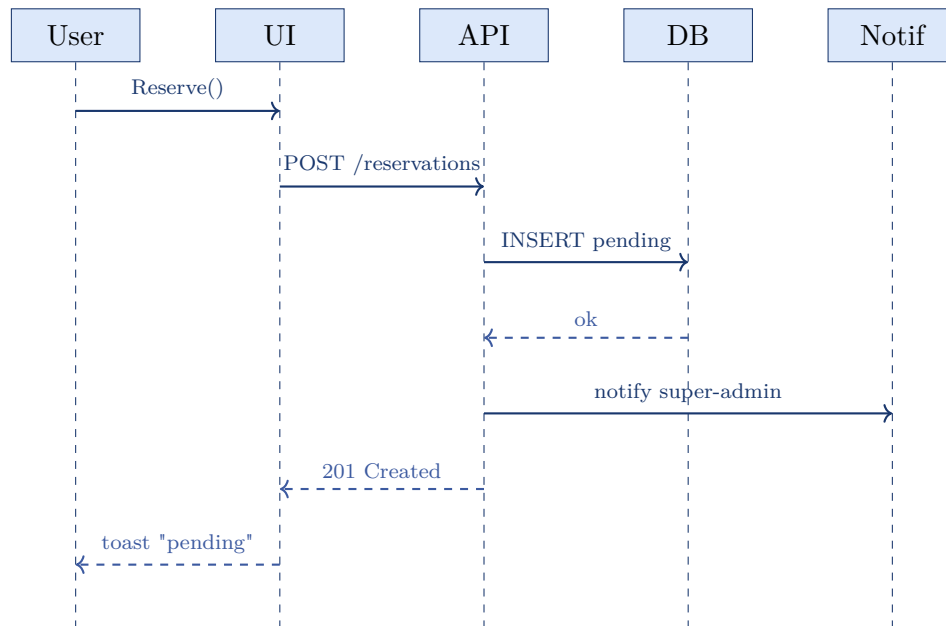


Figure 5.2: Sequence diagram : Reserve and notify

5.5 Class Diagram of the Module

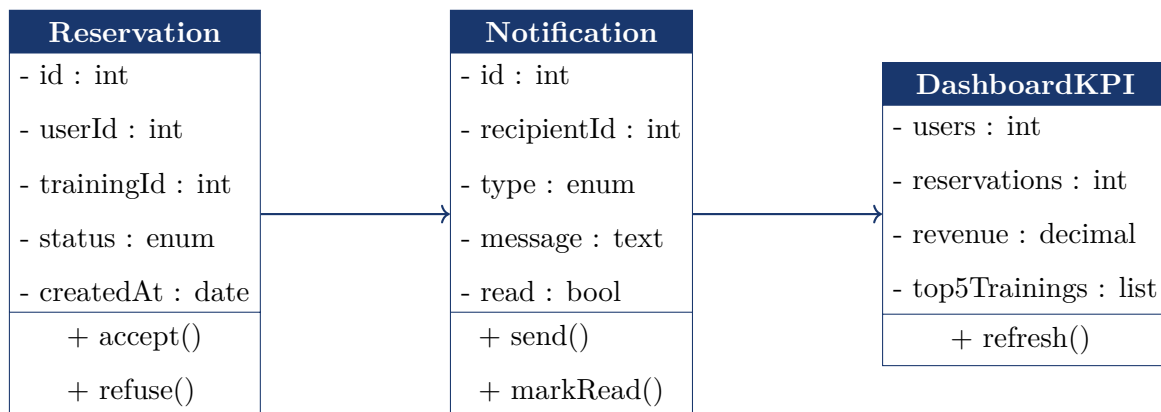


Figure 5.3: Class diagram – Reservation, Notification & Dashboard

5.6 Smart Dashboard & KPIs

The smart dashboard, accessible to administrators and the super-administrator, exposes the platform's pulse in real time.

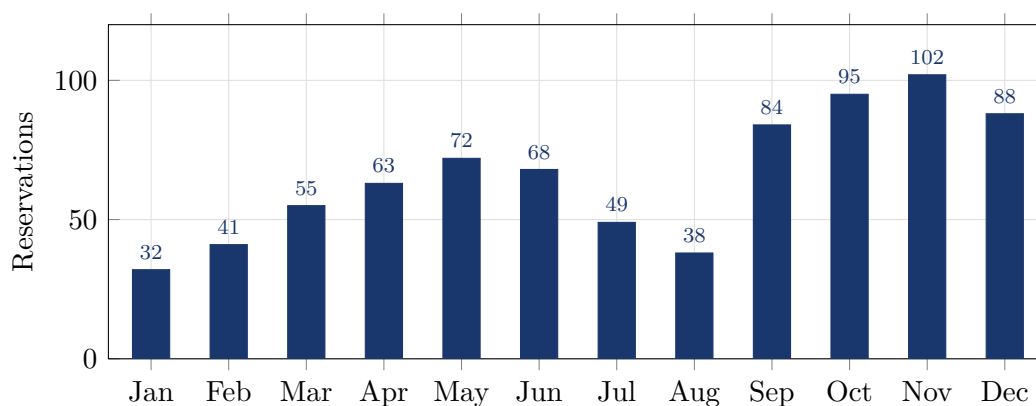


Figure 5.4: Monthly reservations over a representative year (sample data)

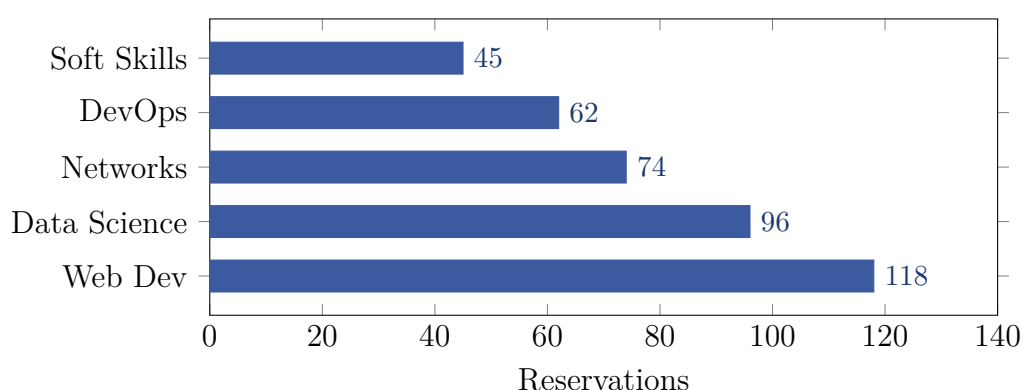


Figure 5.5: Top 5 trainings by reservations (sample data)

5.6.1 Chatbot

The integrated chatbot answers natural-language queries from the admin (e.g. *"how many reservations this week?"*, *"top 3 trainings"*) by routing them to the dashboard service. It enables a faster decision loop without writing any SQL or navigating multiple pages.

5.7 Achieved Increment

- Full reservation workflow with pending / accepted / refused statuses.
- Real-time notifications (in-app + e-mail) for users, admins and super-admin.
- Smart dashboard with KPIs, charts and quick filters.
- Conversational chatbot that exposes the dashboard in natural language.
- Excel/PDF export and Excel import for users and reservations.

5.8 Conclusion

With this third release, the ADVANCIA platform reaches its target perimeter. The next chapter steps back from features and looks at the *realization* : the technical environment, the architecture and the most representative interfaces of the final product.

Chapter 6 Realization

1. Introduction
2. Hardware Environment
3. Software Environment
4. Adopted Architecture
5. Database Schema
6. Project Statistics
7. Representative Interfaces
8. Tests and Validation
9. Conclusion

6.1 Introduction

After describing *what* the platform does, this last chapter explains *how* it has been built. We present the hardware and software environments, the adopted architecture, the database schema, a few representative interfaces and the validation strategy that supports the quality of the final product.

6.2 Hardware Environment

Component	Specification
Processor	Intel Core i5 / i7 (10th generation or higher)
RAM	8 – 16 GB DDR4
Storage	256 – 512 GB SSD
Display	14" – 15.6" Full HD
Network	Wi-Fi 5 / Wi-Fi 6, Ethernet

Table 6.1: Hardware environment used during development

6.3 Software Environment

6.3.1 Languages and frameworks

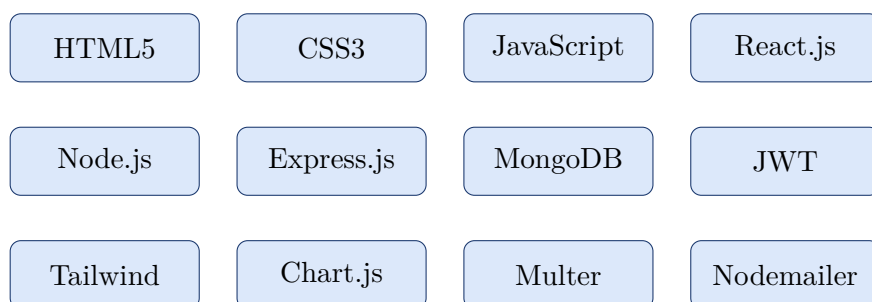


Figure 6.1: Technical stack of the ADVANCIA platform

6.3.2 Tools

Tool	Usage
Visual Studio Code	Main IDE for front-end and back-end development.
Git & GitHub	Source-code management and collaboration.
Postman	Manual testing of the REST API.
Figma	UI/UX design and prototyping.
Draw.io / StarUML	UML modeling.
Trello / Jira	Sprint and backlog tracking.
MongoDB Compass	Database visualization.
L ^A T _E X (Overleaf)	Writing of the present dissertation.

Table 6.2: Tools used during the project

6.4 Adopted Architecture

The platform follows a classical three-tier architecture, with a clear separation between the presentation layer (React SPA), the business layer (Express REST API) and the data layer (MongoDB).

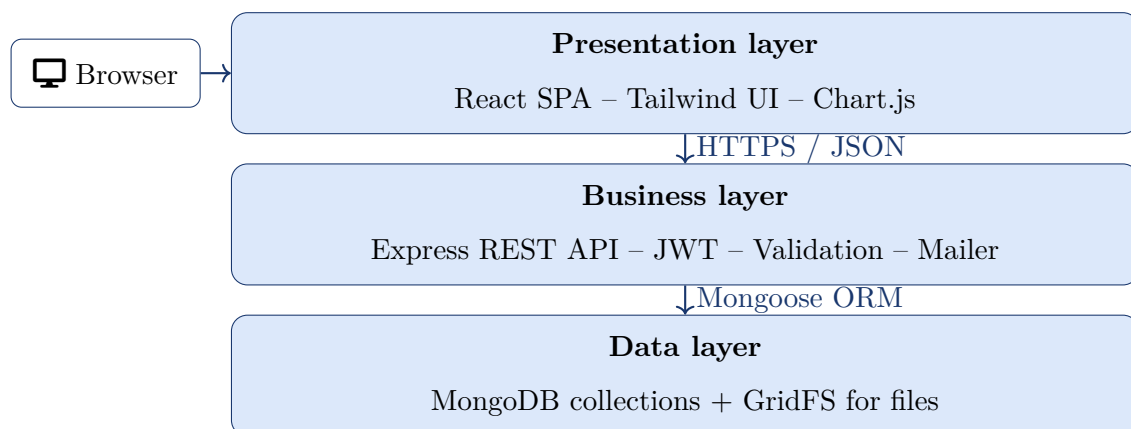


Figure 6.2: Three-tier architecture of the ADVANCIA platform

6.4.1 Deployment view

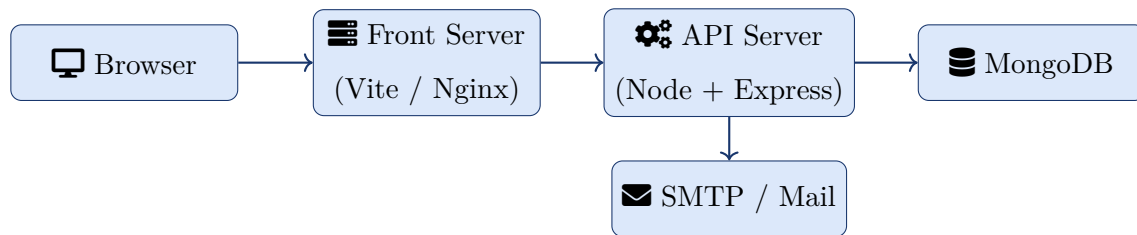


Figure 6.3: Simplified deployment diagram

6.5 Database Schema

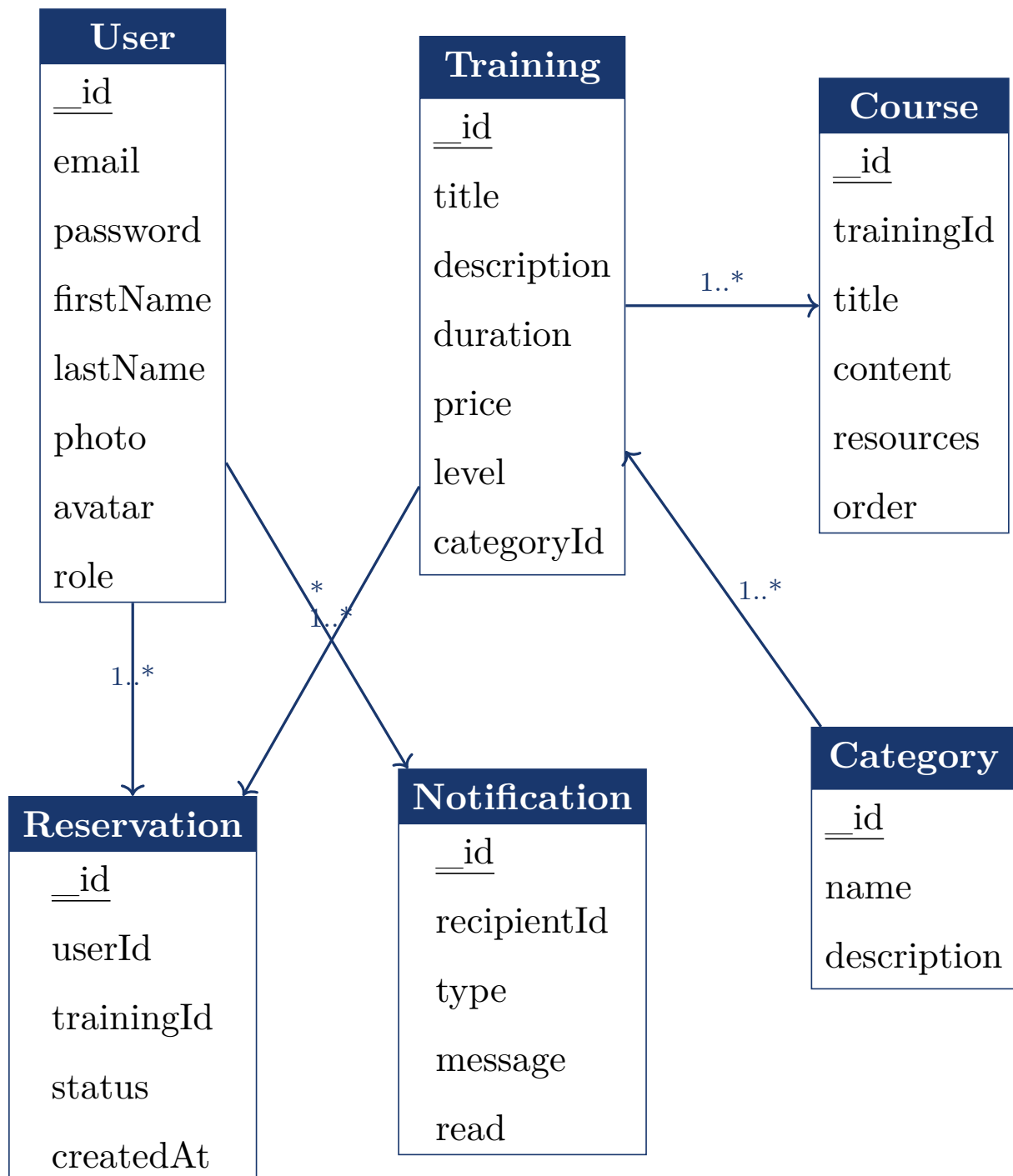


Figure 6.4: Entity-relationship schema of the ADVANCIA database

6.6 Project Statistics

Indicator	Value
Number of user stories	18
Number of releases	3
Number of sprints	6
Total story points	95
Front-end pages	~ 22
REST API endpoints	~ 45
Database collections	7
Manual test cases	~ 60

Table 6.3: Quantitative indicators of the project

6.6.1 Code distribution

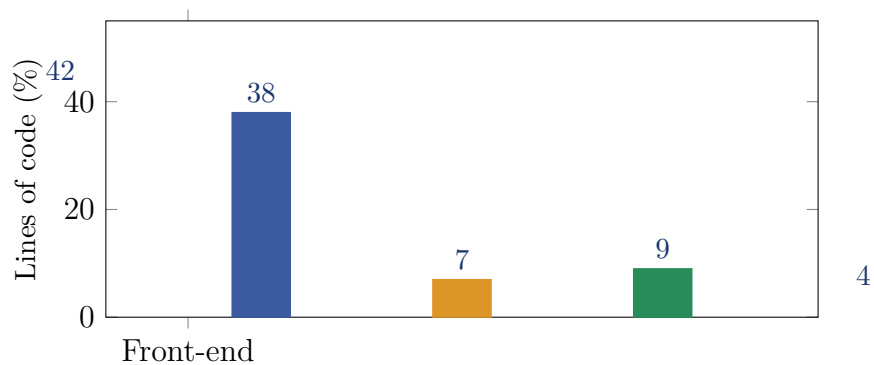


Figure 6.5: Approximate distribution of the code base

6.6.2 Sprint velocity

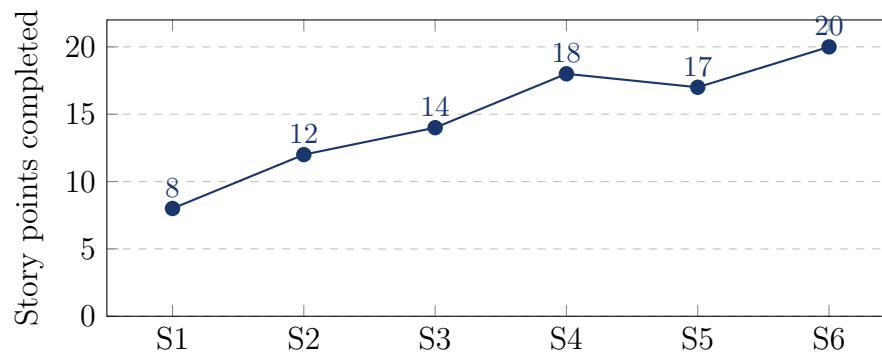


Figure 6.6: Velocity per sprint (story points completed)

6.7 Representative Interfaces

This section illustrates the main interfaces produced. Real screenshots will replace the placeholders in the final printed version of the dissertation.

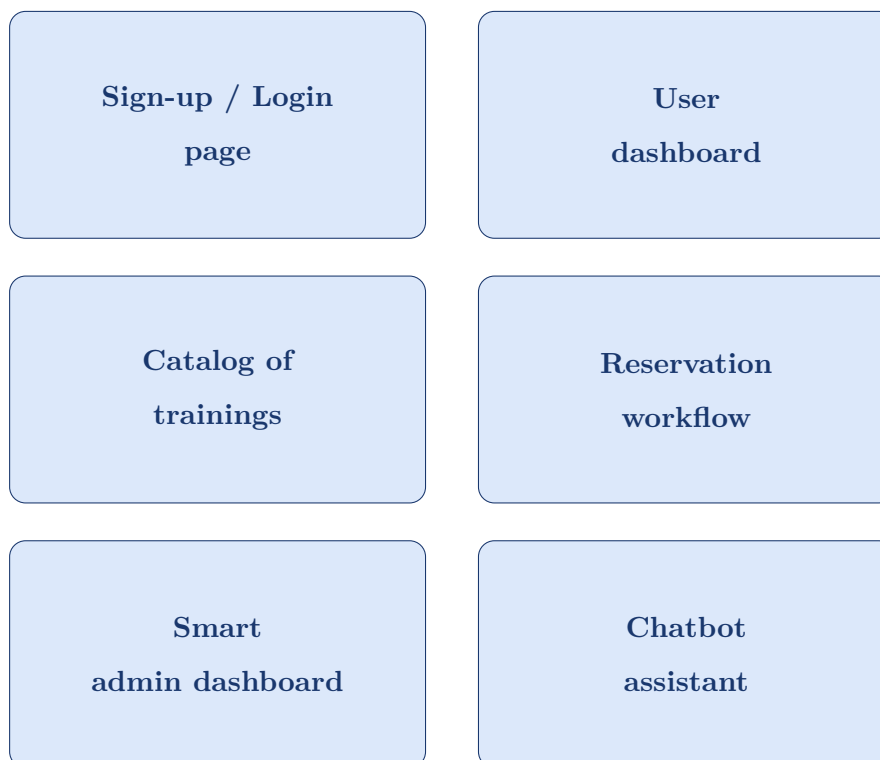


Figure 6.7: Mosaic of the main interfaces of the platform

6.8 Tests and Validation

A pyramid testing strategy was followed during the project : unit tests at the bottom (most numerous, fastest), integration tests in the middle, and end-to-end tests at the top (less numerous, slower but representative of the user journey).

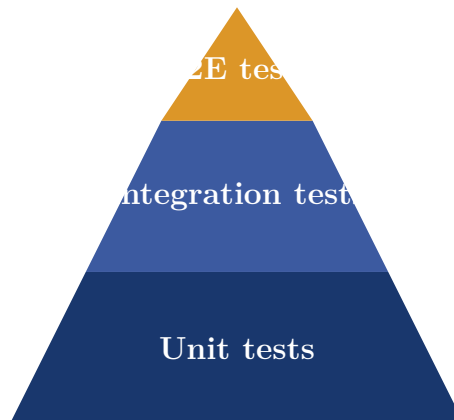


Figure 6.8: Testing pyramid adopted

Test	Cases	Pass rate	Coverage
Unit (services)	38	100 %	Auth, reservation, dashboard
Integration (API)	16	100 %	Critical endpoints
E2E (UI)	8	100 %	Sign-up, reserve, accept, export

Table 6.4: Synthesis of the validation tests

6.9 Conclusion

This realization chapter has materialized the project on a technical level : a modern stack, a clean three-tier architecture, a documented database, measurable indicators and a structured testing strategy. The platform is now ready for delivery and for the perspectives of evolution detailed in the general conclusion.

General Conclusion

The journey we have just described is, at its heart, the story of a problem turned into a solution. Faced with a daily reality where reservations, user records, communications and reports were scattered across paper, spreadsheets and phone calls, our mission was to bring them together inside a single, modern and intelligent web platform : **ADVANCIA**.

Throughout this dissertation, we have followed a clear thread. The *first chapter* settled the framework of the project : the host company ADVANCIA TRAINING CENTER, the limitations of the existing solutions and the chosen agile process. The *second chapter* translated the need into a precise plan – four actors, eighteen user stories, three releases and a Gantt schedule. *Chapters three to five* delivered the platform, sprint after sprint : authentication and user management, training catalog and course management, then reservations, notifications, smart dashboard and chatbot. The *sixth chapter* closed the loop by anchoring the product on a clean technical environment and a measured quality strategy.

The result is a coherent platform, validated against its initial objectives :

- A **secure identity layer** based on JWT and role-based access control.
- A **four-actor model** – User, Trainer, Admin, Super-Admin – each with a tailored experience.
- A **complete reservation workflow**, from request to acceptance, with traceable status transitions.
- A **smart dashboard** backed by real KPIs and supported by an integrated **chatbot**.
- A **multi-format data exchange** (Excel & PDF), aligned with the daily operations of the company.

On a personal level, this project has been far more than a development exercise. It taught us how to listen to a real client, how to translate fuzzy expectations into precise stories,

how to defend technical choices and how to deliver value in short, measurable cycles. It also reinforced our conviction that *good software is, first and foremost, a faithful answer to a human need.*

Perspectives. Several evolutions can naturally extend the present work :

- A **native mobile application** (React Native / Flutter) reusing the existing REST API.
- An **AI-assisted recommendation engine** suggesting personalized trainings to each user.
- An **online payment** module to close the reservation cycle end-to-end.
- A more advanced **learning analytics** layer for trainers (engagement, completion, retention).
- A **multilingual interface** (Arabic / French / English) to broaden the audience.

We hope that **ADVANCIA** will, beyond this dissertation, find its place as a useful tool for its users and as a stepping stone for the future engineers who will continue to extend it.

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